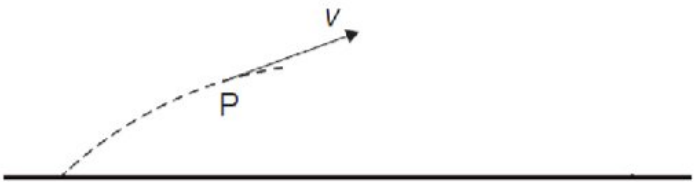
10 The diagram shows the trajectory of a projectile and the velocity *v* of the projectile at point P in its trajectory. P is located before the projectile reaches the peak altitude. Air resistance acts on the projectile. The acceleration of the projectile at P is *a*.



What are the magnitudes of the horizontal component and the vertical component of the acceleration of the projectile at P?

	Horizontal component of *a*	**Vertical component of *a***
A	zero	greater than 9.8 m s^{-2}
B	non-zero	greater than 9.8 m s^{-2}
C	zero	9.8 m s^{-2}
D	non-zero	9.8 m s^{-2}